SAUCEDEMO

Functional Test Case Catalogue

Automation-ready test design for Selenium WebDriver + Java + TestNG/JUnit

Application: https://www.saucedemo.com/ **Scope:** Login, inventory, sorting, cart, checkout, navigation and session behaviour

Purpose: Use this catalogue as the test-design baseline for building a maintainable UI automation suite. Each case includes a clear expected result and a suggested automation priority.

Important: SauceDemo is a demo application. UI behaviour and test data can change. Confirm each expected result against the live application before treating the case as a release acceptance test. This document covers the standard public SauceDemo shopping flow and common test variations; it is not a guarantee that every possible defect or browser/device combination is covered.

Recommended execution order

1. Smoke: valid login → inventory → add item → cart → checkout → finish.
2. Core regression: login negatives, product details, sorting, cart quantity/removal, checkout validations.
3. Extended: menu/session behaviour, alternate users, responsive layout, browser compatibility and accessibility checks.

Prepared as a practical learning and implementation guide. Use stable locators, explicit waits, reusable page objects, isolated test data and independent test setup.

1. Login and authentication

Use standard_user for the main happy path. SauceDemo also provides special-purpose users; treat their unusual behaviour as intentional demo coverage, not necessarily a production defect.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-001	Open login page	Navigate to the base URL.	Login page loads; username, password and Login control are visible.	P0	Yes
SD-002	Login with standard user	Enter standard_user / secret_sauce; click Login.	User is authenticated and inventory page is displayed.	P0	Yes
SD-003	Login with invalid username	Enter an unknown username and valid-looking password.	Login is rejected and a clear error message is displayed.	P1	Yes
SD-004	Login with invalid password	Enter standard_user and an incorrect password.	Login is rejected; user remains on login page.	P1	Yes
SD-005	Login with blank credentials	Leave both fields empty; click Login.	Validation message requests a username.	P1	Yes
SD-006	Login with blank password	Enter username only; click Login.	Validation message requests a password.	P1	Yes
SD-007	Login with blank username	Enter password only; click Login.	Validation message requests a username.	P1	Yes
SD-008	Locked-out user	Login using locked_out_user / secret_sauce.	Login is blocked and locked-out message is displayed.	P1	Yes
SD-009	Problem user login	Login using problem_user / secret_sauce.	Login outcome is observed; record any broken/mismatched inventory UI as a known demo-user behaviour.	P2	Yes
SD-010	Performance glitch user	Login using performance_glitch_user / secret_sauce.	Login eventually completes; waits are resilient to delayed page response.	P2	Yes
SD-011	Password masking	Type a password on login page.	Password characters are masked.	P2	Yes
SD-012	Logout from inventory	Login, open menu, select Logout.	Session ends and login page is displayed.	P1	Yes

2. Inventory, product details and sorting

Validate UI state and data consistency. For sorting, collect displayed names/prices and compare them with a sorted copy rather than checking only the first item.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-013	Inventory page content	Login as standard_user.	Product list, product names, prices, images and Add to cart controls are visible.	P0	Yes
SD-014	Open product details	Select a product name or image.	Correct product detail page opens with matching name, description, price and image.	P1	Yes
SD-015	Return from product details	Open a product and select Back to products.	Inventory page is displayed again.	P1	Yes
SD-016	Add one product	Add one selected product to cart.	Button changes to Remove; cart badge shows 1.	P0	Yes
SD-017	Add multiple products	Add two or more distinct products.	Cart badge equals number of selected products; each button changes to Remove.	P0	Yes
SD-018	Remove product from inventory	Add a product, then select Remove on inventory page.	Product is removed from cart; badge updates or disappears when empty.	P1	Yes
SD-019	Sort by name A to Z	Choose Name (A to Z).	Visible product names are in ascending alphabetical order.	P1	Yes
SD-020	Sort by name Z to A	Choose Name (Z to A).	Visible product names are in descending alphabetical order.	P1	Yes
SD-021	Sort by price low to high	Choose Price (low to high).	Visible prices are in ascending numeric order.	P1	Yes
SD-022	Sort by price high to low	Choose Price (high to low).	Visible prices are in descending numeric order.	P1	Yes
SD-023	Sort after adding item	Add a product, change sort order.	Sort is applied and cart contents remain intact.	P2	Yes
SD-024	Inventory cart navigation	Add item and click cart icon.	Cart page opens and selected item is present.	P0	Yes

3. Shopping cart

Check both the cart badge and the actual cart contents. Keep tests independent by clearing application state before each test.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-025	Open empty cart	Login and open cart without adding products.	Cart page opens with no cart items; checkout cannot proceed with an item that is not present.	P1	Yes
SD-026	Verify cart item details	Add a product and open cart.	Correct product name, description, price and quantity are displayed.	P0	Yes
SD-027	Remove item from cart	Add product, open cart, click Remove.	Item disappears and cart badge updates.	P0	Yes
SD-028	Continue shopping	Open cart and click Continue Shopping.	Inventory page opens; existing cart contents are retained.	P1	Yes
SD-029	Multiple cart items	Add at least three different products and open cart.	All selected products appear exactly once with correct details.	P1	Yes
SD-030	Cart state across navigation	Add item, navigate to details and back to cart.	Cart item and badge remain consistent during navigation.	P1	Yes

4. Checkout and order completion

Use a known product set so totals can be calculated. Assert displayed subtotal and tax using the application's displayed values; avoid hard-coding tax assumptions if the demo changes.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-031	Start checkout with item	Add item, open cart, click Checkout.	Checkout information page opens.	P0	Yes
SD-032	Checkout with valid details	Enter first name, last name and postal code; click Continue.	Checkout overview page opens.	P0	Yes
SD-033	Checkout with all fields blank	Leave all fields empty; click Continue.	Validation message identifies required first name.	P1	Yes
SD-034	Checkout missing first name	Enter last name and postal code only.	Validation message requests first name.	P1	Yes
SD-035	Checkout missing last name	Enter first name and postal code only.	Validation message requests last name.	P1	Yes
SD-036	Checkout missing postal code	Enter first and last name only.	Validation message requests postal code.	P1	Yes
SD-037	Checkout special characters	Enter permitted punctuation in name fields and a postal code.	Application handles input without crashing; validation follows observed rules.	P2	Yes
SD-038	Checkout long input	Enter long values in each field and continue.	Fields handle input safely; page remains usable and validation is consistent.	P2	Partial
SD-039	Cancel checkout information	Open checkout information and click Cancel.	Cart page opens; cart contents remain available.	P1	Yes
SD-040	Verify overview line items	Complete customer information.	Overview shows the same products, quantities and prices as cart.	P0	Yes
SD-041	Verify item total	Proceed to overview with known cart items.	Item total equals sum of item prices, excluding tax.	P1	Yes
SD-042	Verify tax and total	Proceed to overview.	Tax is displayed and order total equals item total plus tax, using displayed values.	P1	Yes
SD-043	Cancel from overview	On overview page click Cancel.	User returns to inventory page; order is not completed.	P1	Yes
SD-044	Finish purchase	On overview page click Finish.	Order confirmation page is displayed with successful completion message.	P0	Yes
SD-045	Return home after purchase	On confirmation page click Back Home.	Inventory page opens; completed checkout flow is exited.	P1	Yes

5. Navigation and session

Menu links and browser navigation may behave differently across browsers. Assert the user-visible outcome and handle additional tabs explicitly where needed.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-046	Open and close side menu	Click menu icon, then close control.	Menu opens and closes without layout or navigation errors.	P1	Yes
SD-047	Menu All Items	Open menu and select All Items.	Inventory page is displayed.	P2	Yes
SD-048	Menu About link	Open menu and select About.	Expected Sauce Labs destination opens in a new/current tab as implemented.	P2	Partial
SD-049	Reset app state	Add products, open menu, select Reset App State.	Cart is cleared and cart badge is removed; verify current page behaviour.	P1	Yes

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-050	Session after refresh	Login and refresh inventory page.	Authenticated session remains usable or behaviour matches the app's session design.	P2	Yes
SD-051	Direct inventory URL while logged out	Log out, then navigate directly to /inventory.html.	Unauthenticated access is prevented or redirected to login.	P1	Yes
SD-052	Browser back and forward	Navigate inventory → details → back; use browser controls.	Navigation remains consistent and no unexpected duplicate/lost cart state occurs.	P2	Partial

6. Extended UI quality and end-to-end coverage

These cases are useful for a broader regression suite. Some are better suited to a manual or cross-browser test pass than a first automation sprint.

ID	Scenario / Test	Steps (summary)	Expected result	Priority	Automation
SD-053	Responsive layout	Open at desktop and narrow mobile viewport sizes.	Controls and product cards remain visible and usable without unintended overlap.	P2	Partial
SD-054	Keyboard navigation	Use Tab, Shift+Tab, Enter and Space through login and shopping flow.	Interactive controls are keyboard reachable and operable in a logical order.	P2	Partial
SD-055	Visible focus state	Tab through key controls.	Focused control has a visible focus indicator.	P2	Partial
SD-056	Image and broken asset check	Inspect inventory and product details.	Product images load; no broken-image placeholders appear for standard user.	P2	Yes
SD-057	Repeated add/remove stability	Repeat add and remove on same item several times.	Cart badge and button label remain synchronized; no duplicate cart entries.	P2	Yes
SD-058	Multi-item checkout end-to-end	Add multiple products, checkout, verify overview and finish.	All items and totals are consistent; confirmation is displayed.	P0	Yes

7. Test data and automation design

Base URL: https://www.saucedemo.com/
Common demo credentials: standard_user / secret_sauce; locked_out_user / secret_sauce; problem_user / secret_sauce; performance_glitch_user / secret_sauce. Keep credentials in a properties/config file rather than scattering them through test methods.

Suggested test suite structure

```
src/test/java
  tests/
    LoginTests.java
    InventoryTests.java
    CartTests.java
    CheckoutTests.java
    NavigationTests.java
  pages/
    LoginPage.java
    InventoryPage.java
    ProductPage.java
    CartPage.java
    CheckoutInfoPage.java
    CheckoutOverviewPage.java
    CheckoutCompletePage.java
    SideMenuComponent.java
  base/
    BaseTest.java
  utilities/
    ConfigReader.java
    WaitUtils.java
    ScreenshotUtils.java
    TestDataProvider.java
src/test/resources
  config.properties
  testng.xml
```

Implementation rules

• Use Page Object Model: locators and page actions belong in page classes; assertions belong primarily in test classes.

• Prefer stable IDs and meaningful CSS selectors. Avoid brittle absolute XPath and index-based locators.

• Use WebDriverWait / explicit waits for visibility, clickability and URL changes. Avoid Thread.sleep except for a temporary diagnostic.

• Use @BeforeMethod to create a fresh browser and establish a known state; use @AfterMethod to capture failure evidence and quit the driver.

• Keep tests independent. Reset app state or create a fresh session before each test that depends on an empty cart.

• Use DataProvider for valid/invalid login and checkout combinations. Keep test data outside test logic.

• Capture screenshots on failure and attach them to Extent Reports if your framework already supports it.

• Do not let tests depend on execution order. A test should pass when run alone and in the full suite.

• Run smoke tests first, then regression. Tag cases with groups such as smoke, regression, checkout and negative.

Suggested first implementation milestone

Build these 10 cases first: SD-001, SD-002, SD-016, SD-024, SD-026, SD-031, SD-032, SD-040, SD-044 and SD-045. This gives you a complete happy-path framework before adding negative and edge cases.

8. Automation checklist

■ Base URL and browser settings are externalized in config.properties.

■ Driver setup and teardown are centralized in BaseTest.

■ Page classes expose reusable actions and read-only getters.

■ Explicit wait utility is reusable and does not hide failures.

- Test methods use clear names and one main purpose per test.
- Assertions verify meaningful outcomes, not just that a click happened.
- Failure screenshots and logs are captured.
- Extent Report records pass/fail/skip with useful test names.
- TestNG groups and DataProviders are configured.
- Suite runs cleanly from the command line and from the IDE.

Coverage summary

Area	Cases	Main coverage
Login & authentication	12	Valid/invalid login, locked user, password masking, logout
Inventory & sorting	12	Products, details, add/remove, sort options
Cart	6	Empty cart, item details, removal, navigation, persistence
Checkout	15	Required fields, overview, totals, cancellation, completion
Navigation & session	7	Menu, reset, refresh, direct URL, browser navigation
Extended UI / E2E	6	Responsive, keyboard, focus, assets, repeated actions, E2E
Total	58	Core functional + extended UI coverage